Computer Hardware

Elliot Attipoe

Hardware

- Computer hardware consist of the system unit and **peripheral devices.**
- Peripheral device is any external device attached to a computer (system unit).
- Examples of peripheral devices include printer, keyboard, monitor, mouse, scanner, speaker and microphone.

How to check computer specs?

- **Windows:**
  1. Open the Start menu.
  2. Click **Settings**.
  3. Click **System**.
  4. Click **About**.

System Summary

Device specifications

HP ENVY x360 m6 Convertible

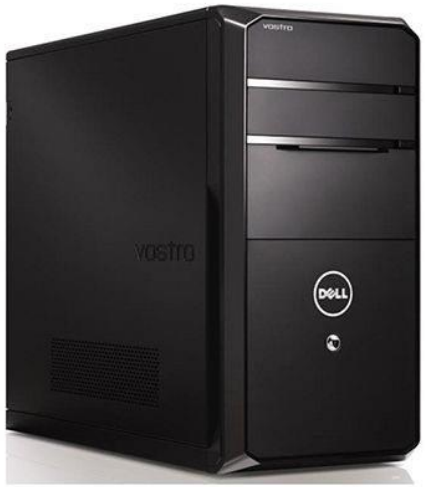
Device name	DESKTOP-MDK01M2
Processor	Intel(R) Core(TM) i7-6560U CPU @ 2.20GHz 2.21 GHz
Installed RAM	12.0 GB (11.9 GB usable)
Device ID	32AF05B4-7DBD-4702-86A0-E039D8BF5455
Product ID	00325-95800-00000-AAOEM
System type	64-bit operating system, x64-based processor
Pen and touch	Touch support with 10 touch points

System Unit

- System unit also known as chassis is the enclosure that contains the main components of a computer.



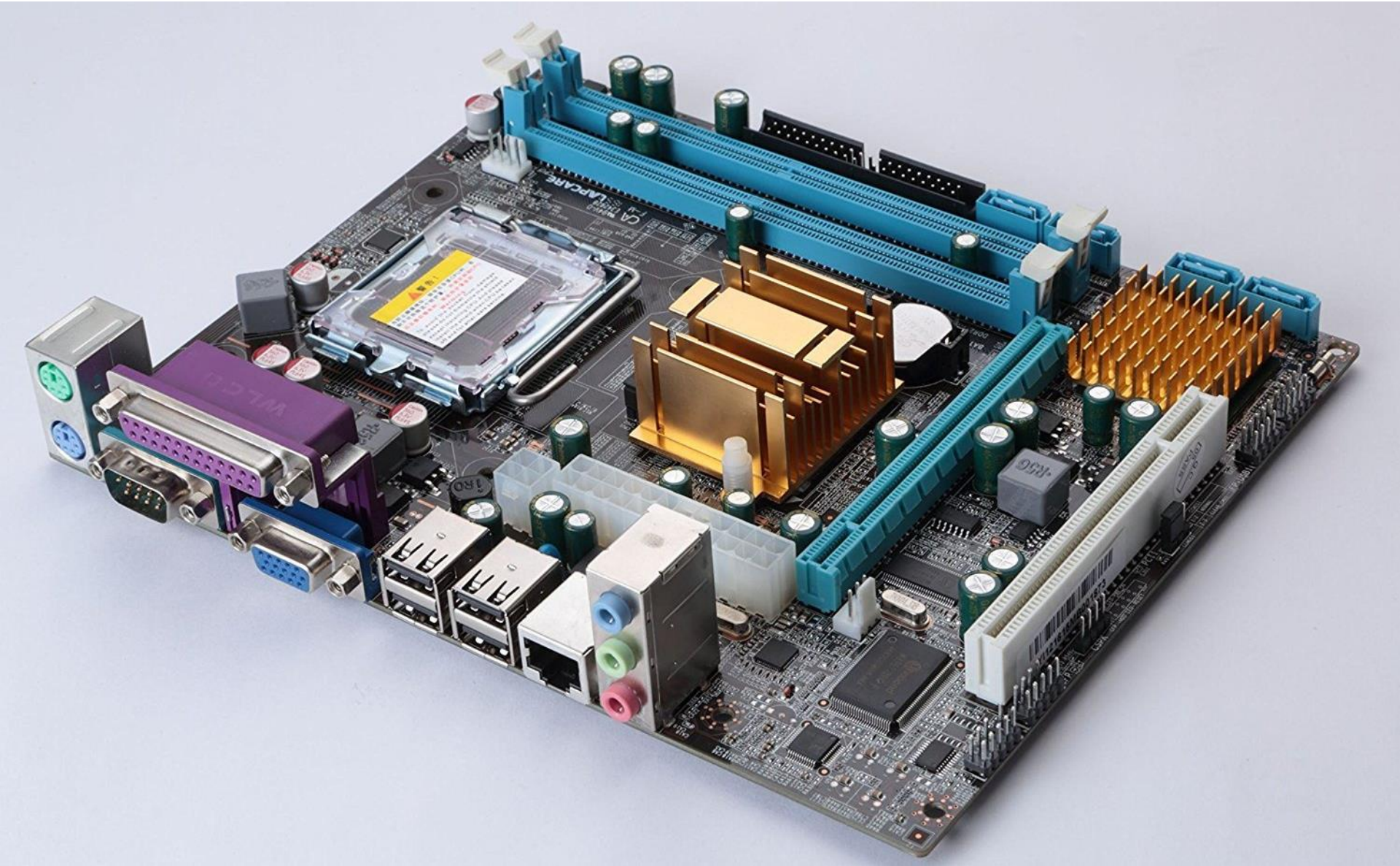
System Unit vs CPU





System Unit VS CPU

Identify the component and label the parts

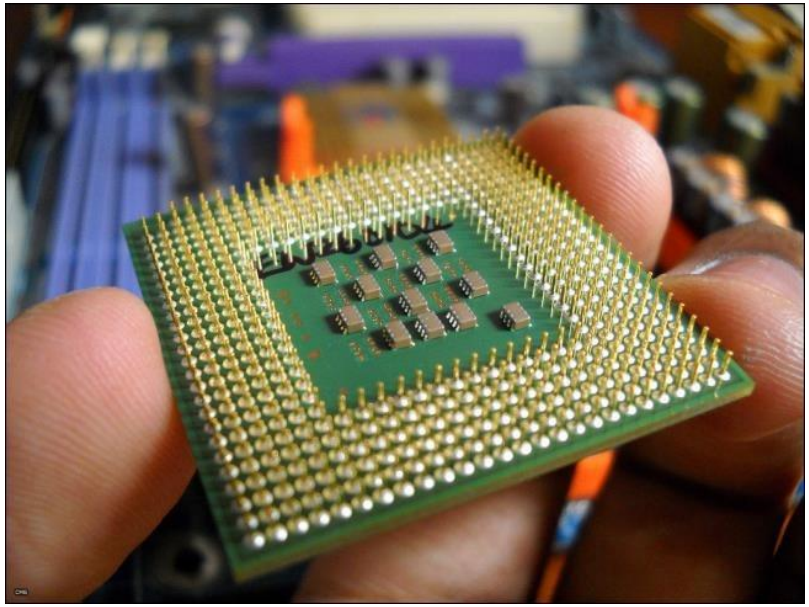


Laptop Motherboard



Central Processing Unit (Processor)





CPU

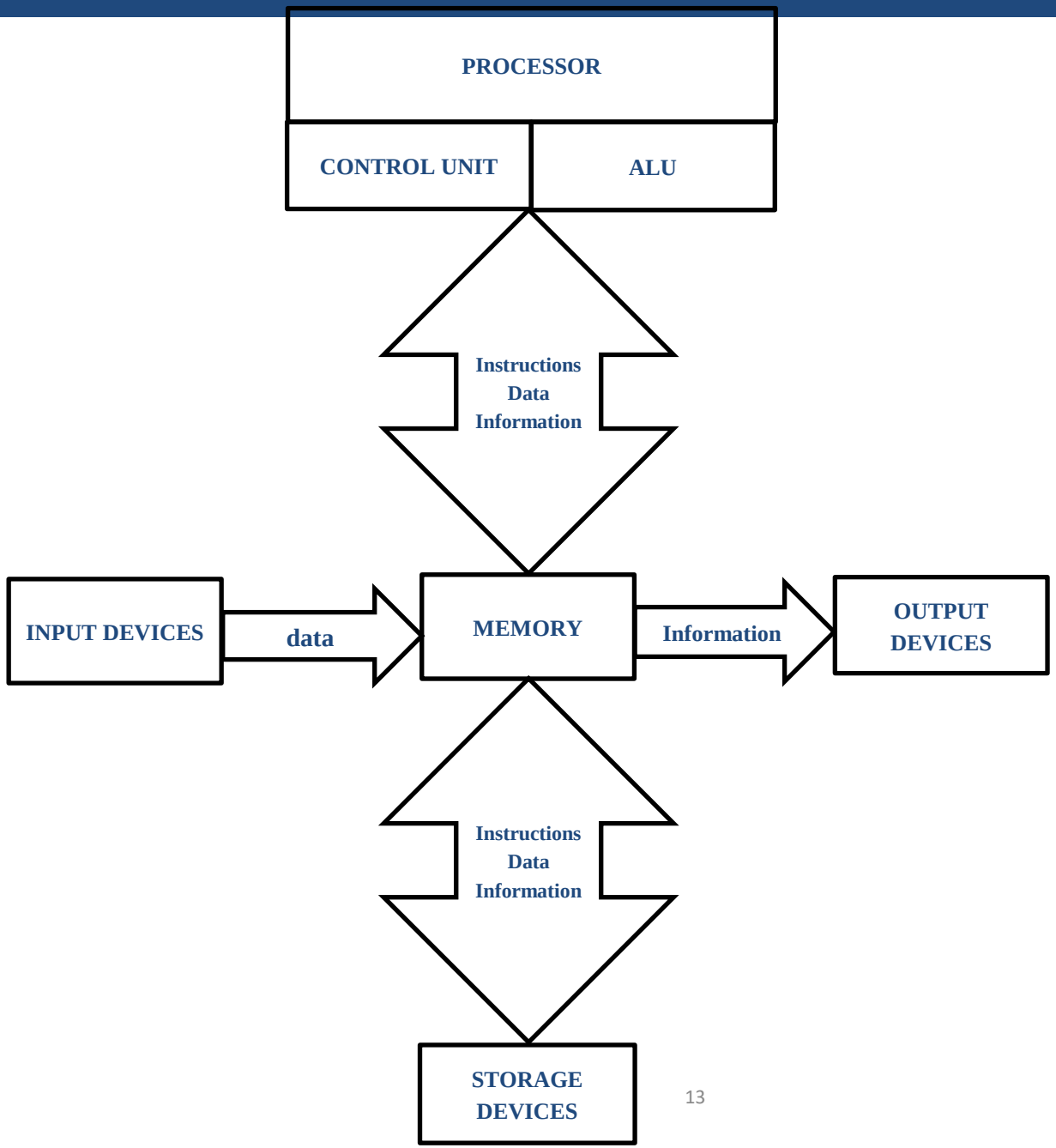


Intel vs AMD

Intel	AMD
Core i9	Ryzen 9
Core i7	Ryzen 7
Core i5	Ryzen 5
Core i3	Ryzen 3

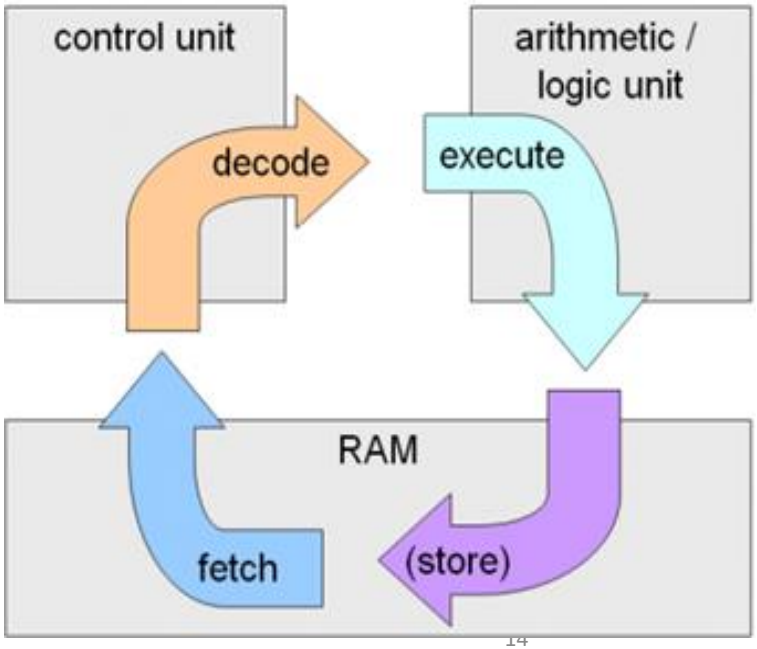
Components of CPU

- The CPU contains:
  - **Control Unit (CU) -** directs and coordinates most of the operations in the computer.
  - **Arithmetic Logic Unit (ALU) -** performs arithmetic, comparison, and other operations. Arithmetic operations include basic calculations, such as addition, subtraction, multiplication, and division.


PROCESSOR
CONTROL UNIT
ALU
Instructions
Data
Information
INPUT DEVICES
data
MEMORY
Information
OUTPUT DEVICES
Instructions
Data
Information
STORAGE DEVICES

Machine Cycle

- A machine cycle consists of the steps that a computer's processor executes whenever it receives an instruction from the RAM or memory.



Machine Cycle

A. **Fetching** is the process of obtaining a program or an application instruction or data item from memory.
B. **Decoding** refers to the process of translating the instruction into signals the computer can execute.
C. **Executing** is the process of carrying out the commands.
D. **Storing** means writing the result to memory (not to a storage medium).

Research

- Differences between registers and cache memory and their relationship to the CPU.

System Clock

- The processor relies on a small quartz crystal circuit called the **system clock** to control the timing of all computer operations.
- The system clock generates regular electronic pulses, or **ticks**.
- Each tick equates to a ***clock cycle***.
- The pace of the system clock, called the **clock speed**, is measured by the number of ticks per second.
- PC processors have clock speeds in the gigahertz range.

System Clock (cont.)

- One **gigahertz (GHz)** equals one billion ticks of the system clock per second.
- A computer that operates at 3 GHz has 3 billion (giga) clock cycles in one second (hertz).
- The faster the clock speed, the more instructions the processor can execute per second.

- **What is a good processor speed?**
- 3.5 to 4.2 GHz

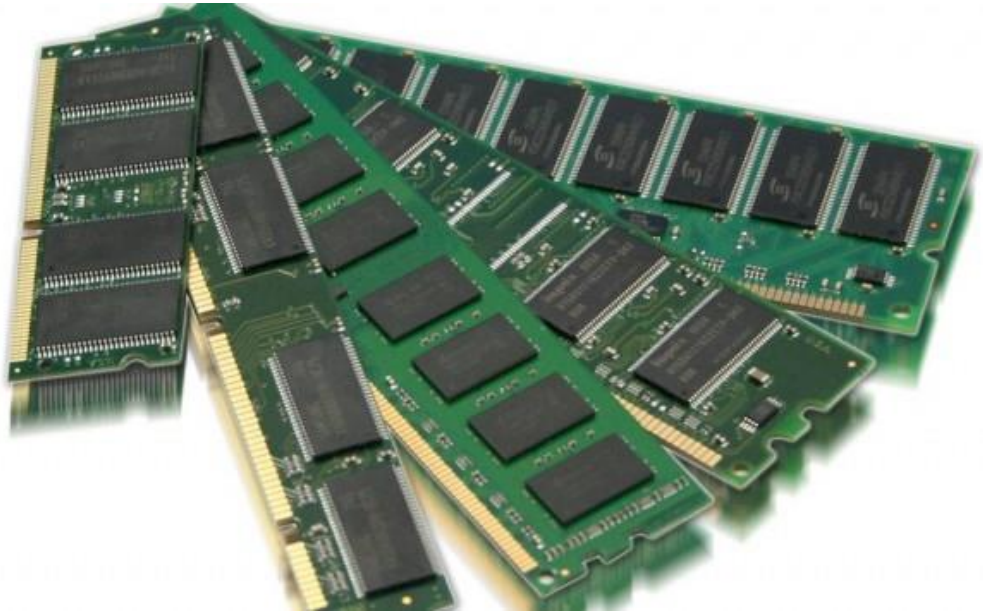
Comparison

No	Processor Model	Processor Speed
1	Intel 11th Generation Core i3	4.1 GHz
2	Intel 12th Generation Core i7 Evo Platform	2.8 GHZ

- Cores – parts of the processors that does parallel processing – aids multi-tasking.
- Each core is a processor technically.

Memory

- ☐ 64 gigabytes
- ☐ 32 gigabytes
- ☐ 16 gigabytes
- ☐ 12 gigabytes
- ☐ 8 gigabytes
- ☐ 6 gigabytes
- ☐ 4 gigabytes
- ☐ 2 gigabytes





Great for business

Dell - Precision 7000 17.3" Laptop - Intel Core i7 - 64 GB Memory - NVIDIA RTX A4000 - 512 GB SSD - Gray

Model: 7JRPT **SKU:** 6477537

☆☆☆☆☆ Not yet reviewed

Unavailable nearby

Pickup: Unavailable within 250 miles of Hato Rey

Update location

Shipping: Unavailable in your area

This item is only available in certain markets.

Estimates for 00820

Compare Save

Price Match Guarantee

$4,199.99

Free 3-month security
A $14.99 value

Add



MSI - WS66 15.6" Gaming Laptop - Intel Core i9 - 64 GB Memory - NVIDIA RTX A5000 - 1 TB SSD - Black

Model: WS6611220 **SKU:** 6469403

★☆☆☆☆ (1)

Unavailable nearby

Pickup: Unavailable within 250 miles of Hato Rey

Update location

Shipping: Unavailable in your area

This item is only available in certain markets.

Estimates for 00820

Price Match Guarantee

$4,199.99

Free 3-month security
A $14.99 value

Add

Memory

- Memory stores three basic categories of items:
  - The operating system and other programs that control or maintain the computer and its devices.
  - Applications that carry out a specific task, such as word processing.
  - The data being processed by the applications and the resulting information.

Types of Memory

- Two main types:
  - –Volatile
  - –Non-Volatile

- Example of volatile memory is RAM
- Example of non-volatile memory is ROM and Flash Memory.

ROM

- The data on most ROM chips cannot be modified — hence, the name read-only.
- What kind of information is stored in the ROM?
- What is firmware?
- **Firmware -** contain permanently written data, instructions, or information, such as a computer or mobile device's start-up instructions.

Flash Memory

- Flash memory is a type of nonvolatile memory that can be erased electronically and rewritten.
- Most computers use flash memory to hold their start-up instructions because it allows the computer to update its contents easily.
- What kind of information is stored on a flash memory?
- Flash memory is the most widely used data storage technology in consumer devices today. It stores our **photos, our videos, our music, our documents** and so much more.

CMOS

- Some RAM chips, flash memory chips, and other memory chips use complementary metal-oxide semiconductor (CMOS pronounced SEE-moss) technology because it provides high speeds and consumes little power.

BIOS

- What is it?
- **Basic Input/Output System,** also known as the **System BIOS, ROM BIOS, BIOS ROM** or **PC BIOS**)
- **BIOS** is firmware used to provide runtime services for operating systems and programs and to perform hardware initialization during the booting process (power-on startup)

Storage

- RAM holds that data temporarily and it is known as **primary storage**.
- For data to be safely and permanently stored, however, it must be placed in **secondary storage**.
- Secondary storage can include hard disk drives, solid state drives, CDs, DVDs, and network and cloud-based storage.

Storage

- Solid State Drive (SSD) Storage
- Hard Disk Drive Storage (HDD)
- Hybrid (SSD + HDD)
- Embedded Multimedia Card (eMMC) Storage



ASUS - GSRF 2-in-1 15.6" 4K Ultra HD Laptop - Intel Core i7 - 16GB Memory - NVIDIA GeForce GTX 1050 - 2TB HDD + 256GB SSD - Gun Metal Gray

Model: GSRF Q536FD-BI7T15 **SKU:** 6453743

(1)

Sold Out

Compare Save

$1,115.99

Save $434 Was $1,

Free 3-month securi
A $14.99 value

Sold O



HP - ENVY 17.3" Touch-Screen Laptop - Intel Core i7-1165G7 - 12GB Memory - 1TB HDD + 256GB SSD

Model: 17-CG1010NR **SKU:** 6462057

(10)

Open-Box Only

None in your area.

Compare Save

$1,159.99

Free 3-month securi
A $14.99 value

Shop Oper

Open-Box: from $985

HDD vs SSD vs eMMC

- Three key factors:
  - –system boot-up times,
  - –application and file loading,
  - –file copy / paste command execution.

Cloud-Based Storage

- A cloud is a secure computing environment consisting of a set of remote servers that users access via the Internet.



Storage Capacities

Unit	Value	Size
Bit (B)	0 or 1	1/8 of a byte
Byte (B)	8 bits	1 byte
Kilobyte (KB)	1000^1 bytes	1,000 bytes
Megabyte (MB)	1000^2 bytes	1,000,000 bytes
Gigabyte (GB)	1000^3 bytes	1,000,000,000 bytes
Terabyte (TB)	1000^4 bytes	1,000,000,000,000 bytes
Petabyte (PB)	1000^5 bytes	1,000,000,000,000,000 bytes
Exabyte (EB)	1000^6 bytes	1,000,000,000,000,000,000 bytes
Zettabyte (ZB)	1000^7 bytes	1,000,000,000,000,000,000,000 bytes
Yottabyte (YB)	1000^8 bytes	1,000,000,000,000,000,000,000,000 bytes

https://seedscientific.com › how-much-data-is-created-e...

How Much Data Is Created Every Day? [27 Staggering Stats]

28 Oct 2021 — Four **petabytes** is the estimated amount of new **data** being **generated** by Facebook **daily**. (Facebook Research, Statista). Instagram's parent company ...

Extra Features - Laptop

Features

- [ ] 2-in-1 Design
- [ ] Backlit Keyboard
- [ ] Bluetooth
- [ ] Built for Windows Ink
- [ ] Built-In Microphone
- [ ] ENERGY STAR Certified
- [ ] EPEAT Qualified
- [ ] Front-Facing Camera
- [ ] Headphone Jack
- [ ] Lightweight/Portable
- [ ] Media Card Reader
- [ ] Microphone Input
- [ ] Non-Touch Screen
- [ ] None
- [ ] Numeric Keypad
- [ ] Rear-Facing Camera
- [ ] Touch Screen
- [ ] Wireless Connectivity
- [ ] Wireless-A
- [ ] Wireless-AC
- [ ] Wireless-AC (867 Mbps)
- [ ] Wireless-AX
- [ ] Wireless-B
- [ ] Wireless-G
- [ ] Wireless-N

Tips to extend Laptop's Battery Life

- Decrease the screen brightness
- Change the power settings – power saver
- Disable WiFi and Bluetooth
- Don't leave your laptop on permanent charge
- Disconnect or remove unnecessary peripherals
- Don't run battery down to empty

Factors that affect speed of a computer

- Processor type and speed
- Memory size
- System type
- Storage type
- USB port type

Input Devices

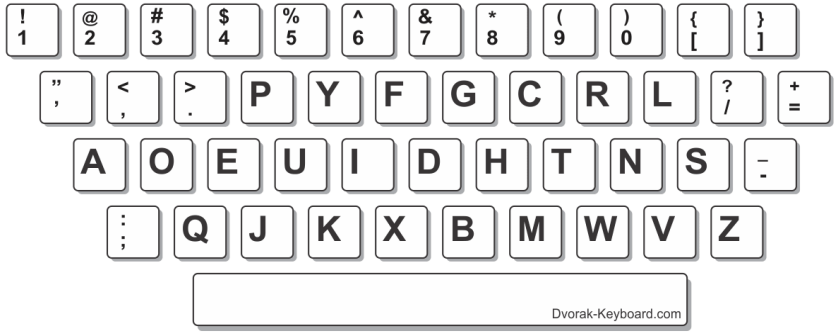
- Input devices allow the user to communicate with the operating system and applications.

Keyboard

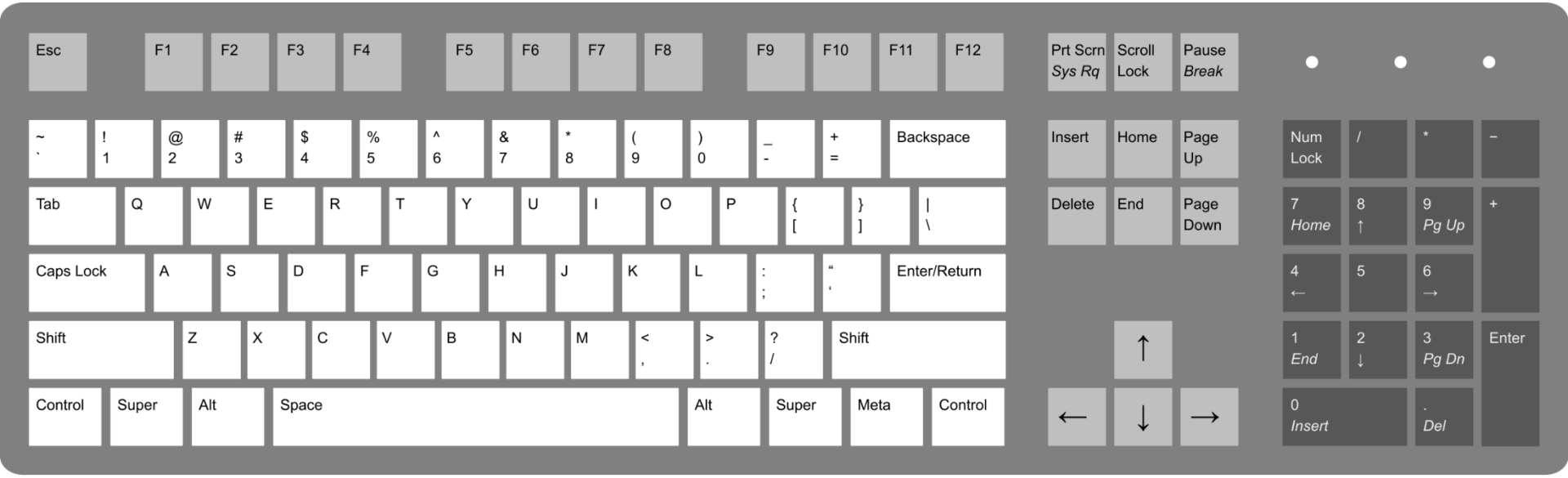
- **QWERTY**



- **Dvorak**

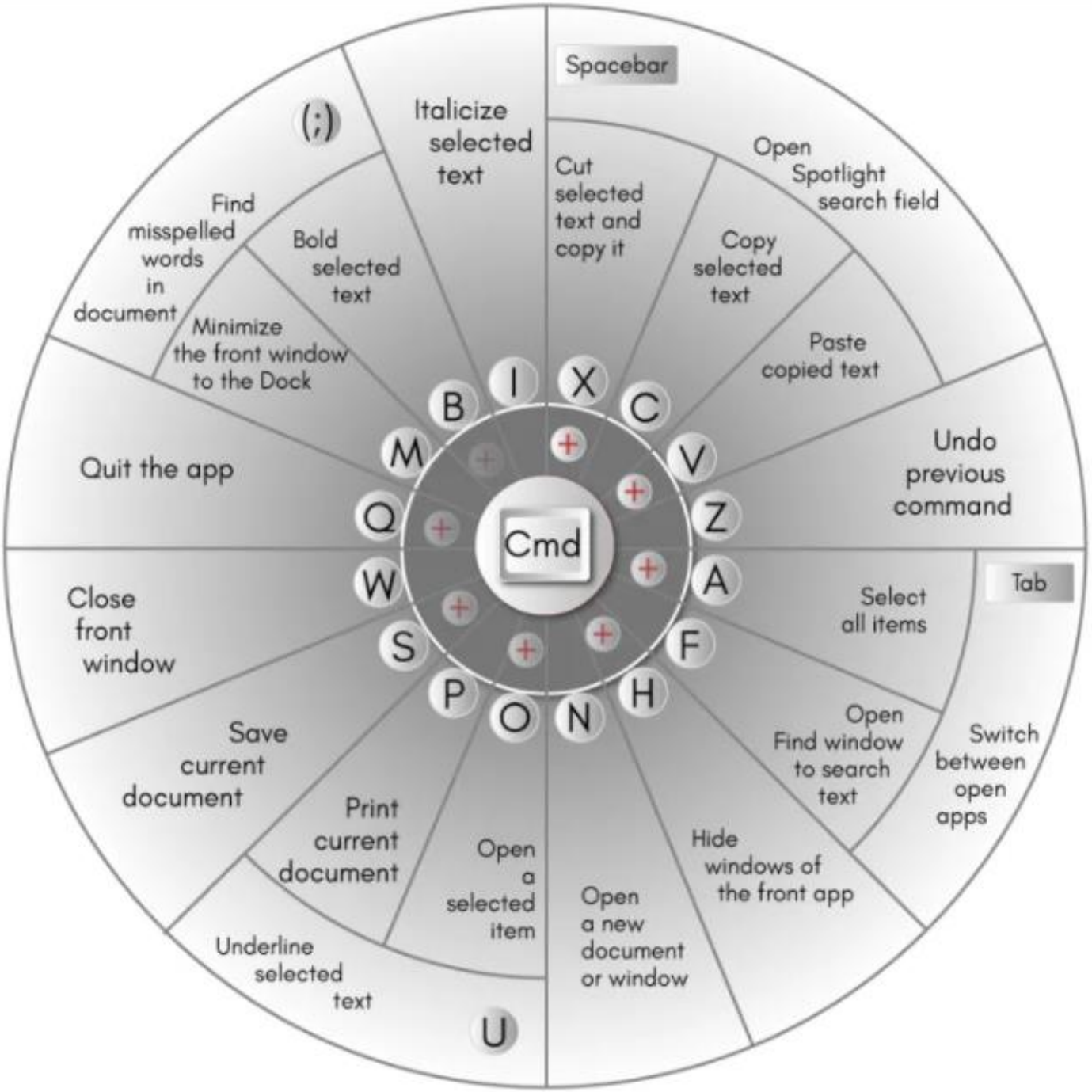


Keyboard Latout



Specialty Keys

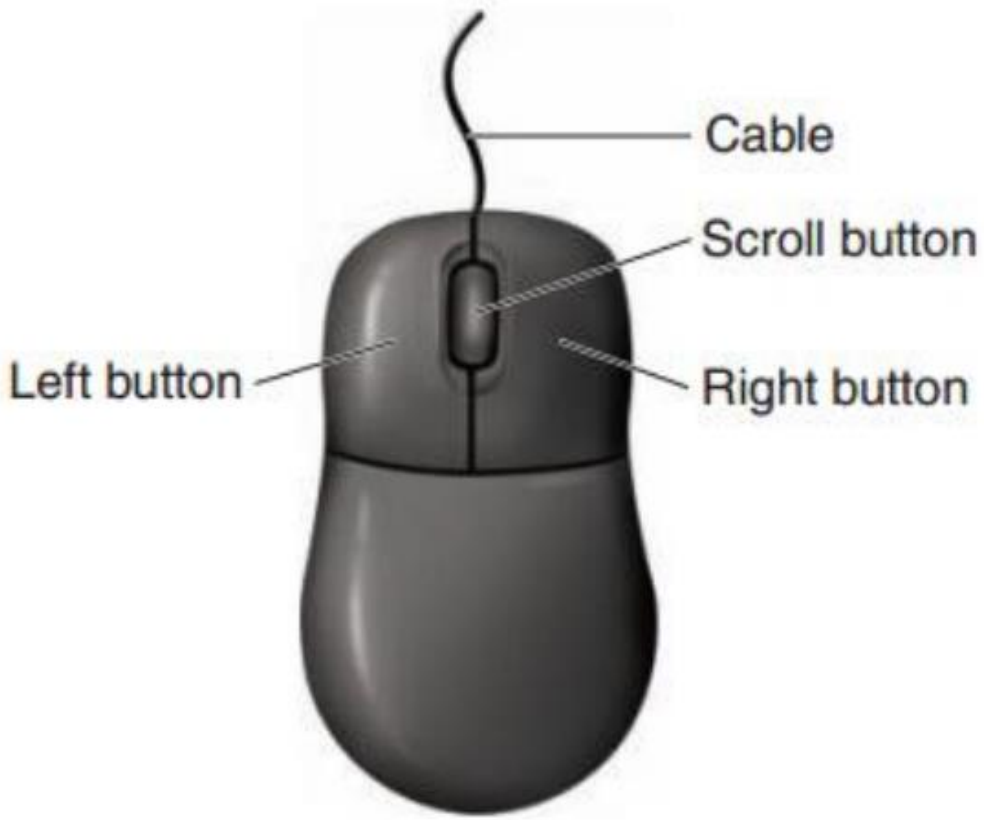
- **Function keys** (F1 through F12) perform special actions when pressed.
- **Toggle keys** turn a feature on or off each time they are pressed.
- **Modifier keys** change the meaning of other keys when they are pressed in combination with them.
- **Positional keys** such as the Home, End, Page Up, and Page Down keys, as well as the directional arrow keys, scroll the display or move the insertion point in applications.


Cmd
Spacebar
Tab
(;)
Italicize selected text
Find misspelled words in document
Bold selected text
Minimize the front window to the Dock
Quit the app
Close front window
Save current document
Print current document
Underline selected text
Open a selected item
Open a new document or window
Hide windows of the front app
Open Find window to search text
Switch between open apps
Select all items
Undo previous command
Paste copied text
Copy selected text
Cut selected text and copy it
Open Spotlight search field
I
B
M
Q
W
S
P
U
O
N
H
F
A
Z
V
C
X

Taking Screenshots of computer screen

- CTRL + PRT SC
- WINDOWS KEY + PRT SC
- ALT + PRT SC

Mouse part







Mouse Operations

- Click
- Right click
- Double Click
- Drag & drop

Mouse Operations

Table 1-1 Mouse Operations

Operation	Description	Common Uses
Point	Move the mouse until the pointer is positioned on the item of choice.	Position the pointer on the screen.
Click	Press and release the primary mouse button, which usually is the left mouse button.	Select or deselect items on the screen or start a program or feature.
Right-click	Press and release the secondary mouse button, which usually is the right mouse button.	Display a shortcut menu.
Double-click	Quickly press and release the primary mouse button twice without moving the mouse.	Start a program or program feature.
Drag	Point to an item, hold down the primary mouse button, move the item to the desired location on the screen, and then release the mouse button.	Move an object from one location to another or draw pictures.

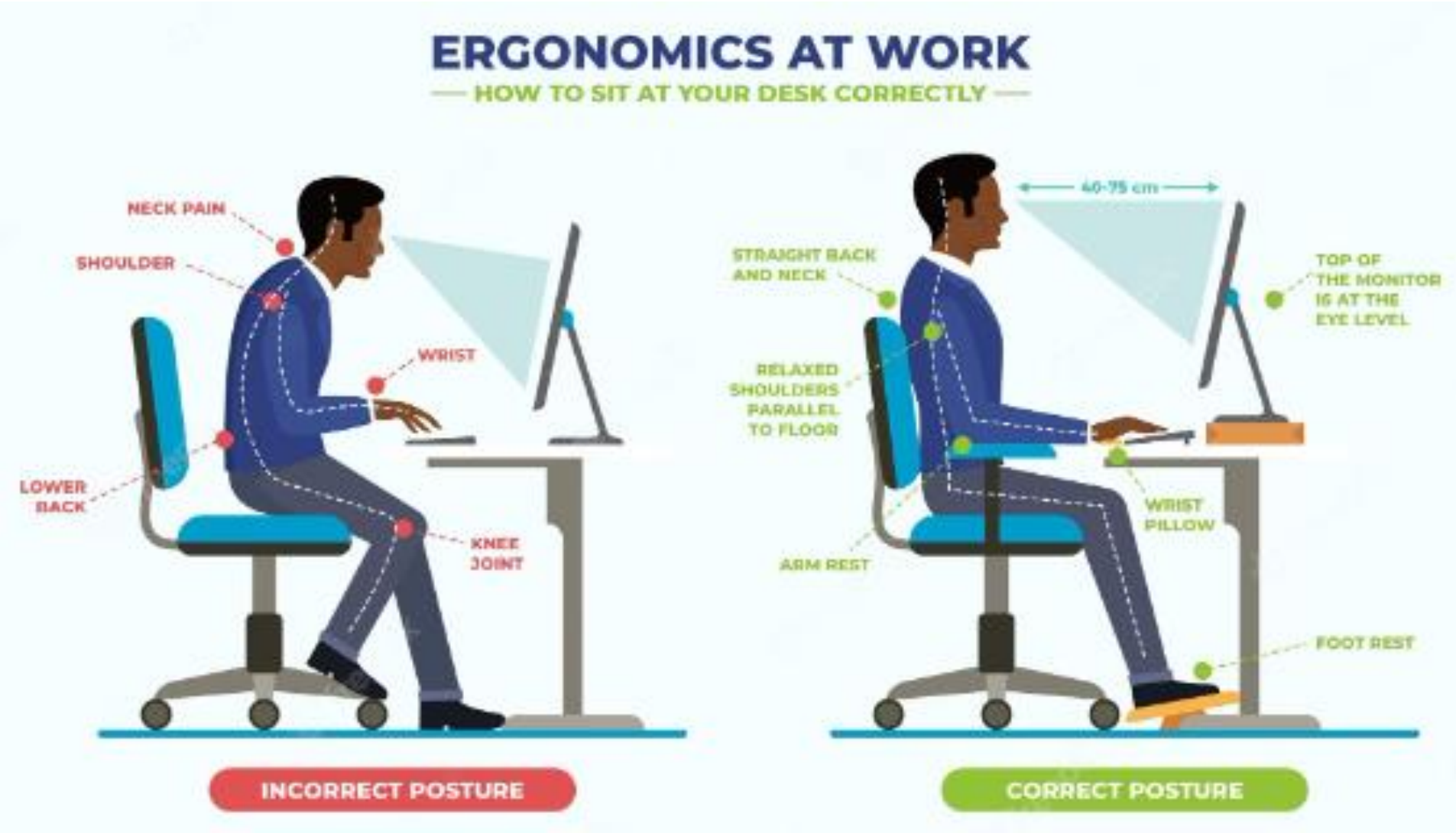
Ergonomics

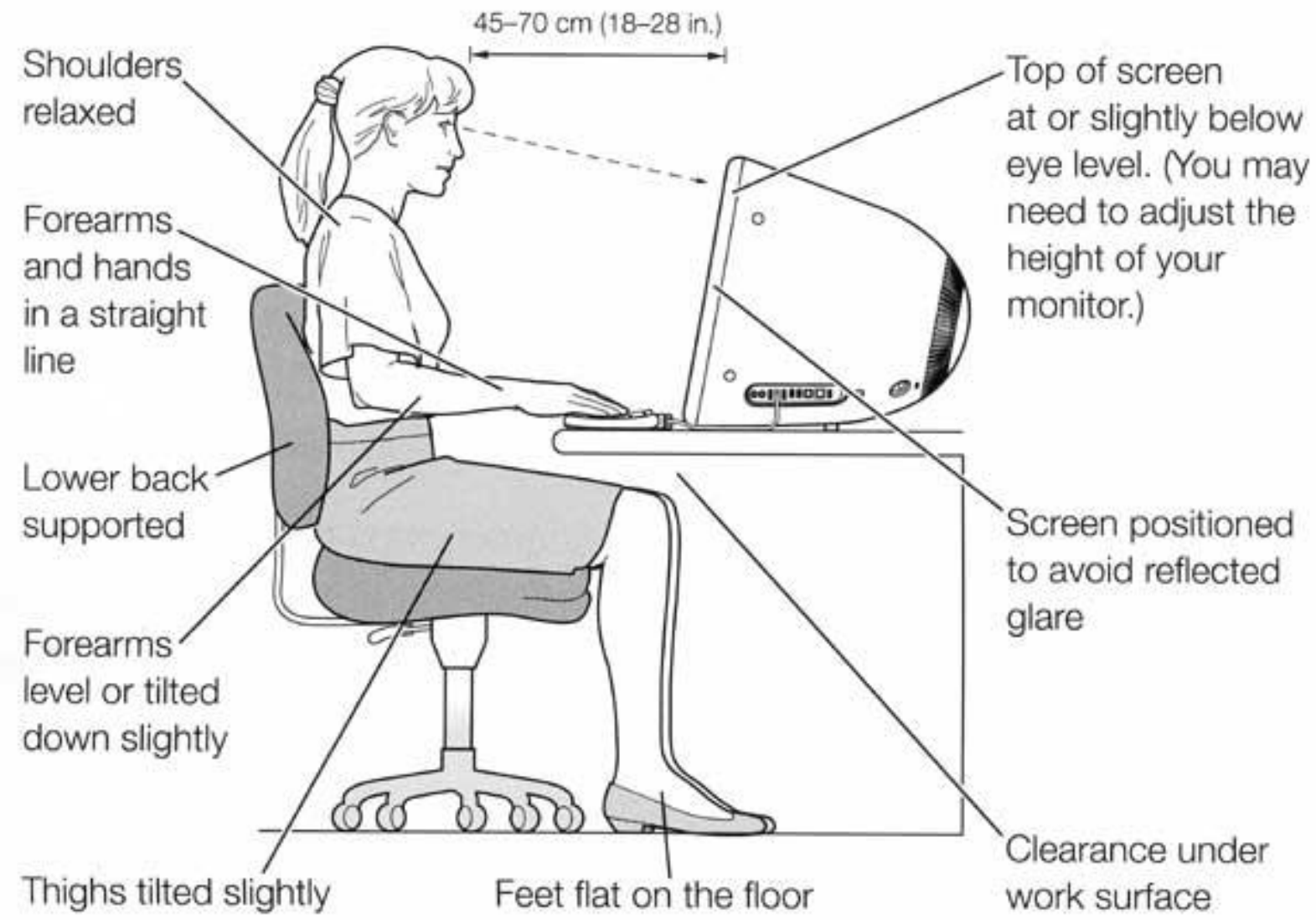
- Ergonomics refers to minimizing injury or discomfort while using the computer.



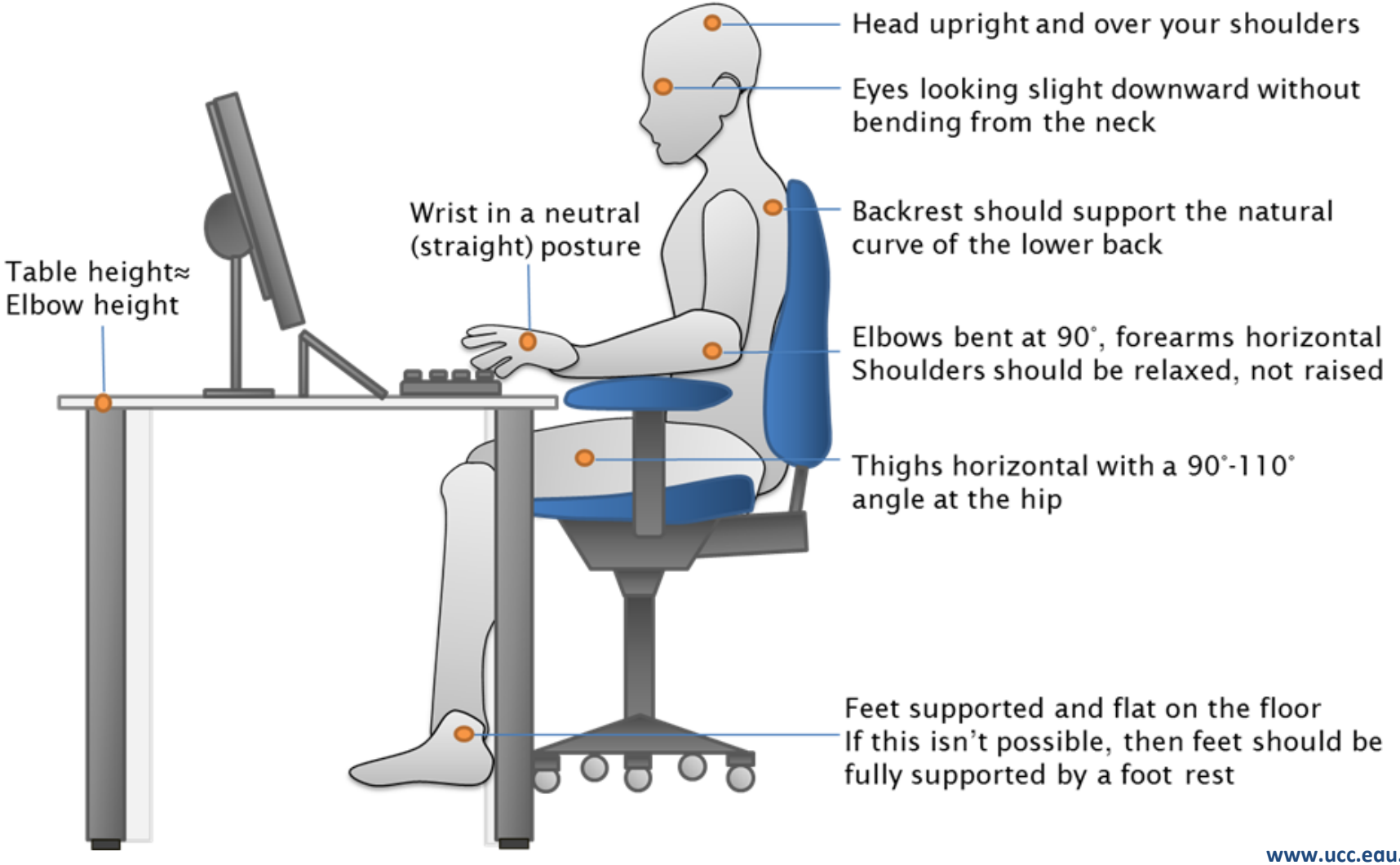





ERGONOMICS AT WORK
HOW TO SIT AT YOUR DESK CORRECTLY
NECK PAIN
SHOULDER
WRIST
LOWER BACK
KNEE JOINT
INCORRECT POSTURE
40-75 cm
STRAIGHT BACK AND NECK
TOP OF THE MONITOR IS AT THE EYE LEVEL
RELAXED SHOULDERS PARALLEL TO FLOOR
WRIST PILLOW
ARM REST
FOOT REST
CORRECT POSTURE


45–70 cm (18–28 in.)
Shoulders relaxed
Forearms and hands in a straight line
Lower back supported
Forearms level or tilted down slightly
Thighs tilted slightly
Feet flat on the floor
Top of screen at or slightly below eye level. (You may need to adjust the height of your monitor.)
Screen positioned to avoid reflected glare
Clearance under work surface

WHAT IS GOOD POSTURE?



Ergonomic Devices







Chair and Workstation Guidelines

- A **workstation** refers to an individual computer, or group of computers used by a single user to perform work.

